

Assignment 7

November 9, 2018

The entirety of the code is shown below. For some of the questions, the output and code are shown concurrently.

```
/* Question 8 */

/* (a) Read the data into SAS and display the mean TotalPoints for each lecture section */

/* Reading the data */
filename myurl url "https://www.uts.utoronto.ca/~butler/c32/SportsExamples.csv";
proc import
    datafile=myurl
    dbms=csv
    out=mydata
    replace;
    getnames=yes;

/* Viewing the data */
proc print;

proc means;
    class section;

/* 8 (b) Obtain a boxplot of total points for each section */

/*
 * There is one categorical variable and one quantifiable variable
 * so it is suitable to make a boxplot
 */
proc sgplot;
    vbox Total_Points / category=section;

/* Question 9 */

/* 9 (a) Read the data into SAS and display the first 20 rows */

filename myurl url "http://www.uts.utoronto.ca/~butler/c32/footprint.txt";
proc import
    datafile=myurl
    dbms=csv
```

```

out=footprint
replace;
getnames=yes;
delimiter=' ';

/* Viewing the data */
proc print data=footprint(obs=20);

/* 9 (b) Make a suitable graph of the ecological footprint */

/* We see that there is one (suitable) categorical variable
 * and one quantitative variable, so a boxplot is used
 */

proc sgplot;
  vbox eco_footprint / category=region;

/* 9 (c) Find the mean and median ecological footprint by region. */
proc means mean median;
  class Region;

```

Question 8

- (a) “Read the data into SAS and display the mean TotalPoints for each lecture section”

The code that was ran to get the output was:

```

filename myurl url "https://www.utsc.utoronto.ca/~butler/c32/SportsExamples.csv";
proc import
  datafile=myurl
  dbms=csv
  out=mydata
  replace;
  getnames=yes;

proc means;
  class section;

```

The output is found as follows

The MEANS Procedure						
Analysis Variable : Total_Points						
Section	N Obs	N	Mean	Std Dev	Minimum	Maximum
Regular	29	29	338.0689655	30.6045410	265.0000000	382.0000000
Sports	28	28	307.2500000	52.5107573	199.0000000	397.0000000

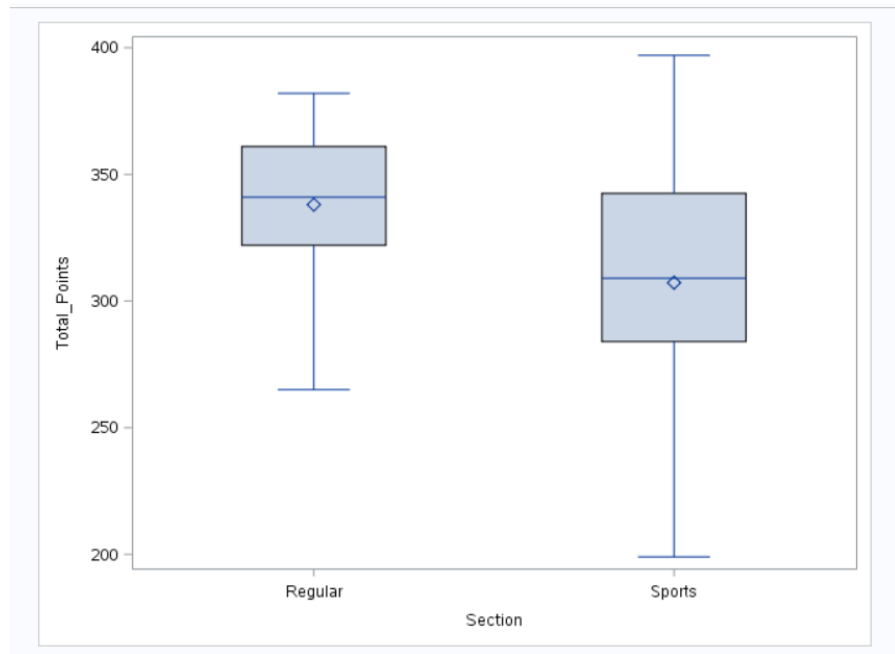
We see that the regular section has a mean of 338.0689655 and the sports section has a mean of 307.25

- (b) “Obtain a boxplot of total points for each section. Does one of the sections have a clearly higher average than the other? Explain briefly.”

My code for this question (bouncing off of (a)) was:

```
proc sgplot;  
vbox Total_Points / category=section;
```

The output for the boxplot is found as follows:



It appears that the Regular section has a higher average in comparison to the Sports section (as this is indicated by that diamond in the boxplot and the means in (a)). A possible reason of such is that the people who are enrolled in the sports section may believe that the Sports section is easier as opposed to the regular section, whereas it was the material is equivalently the same.

- (c) “What is it about the design of this study that would make you hesitant to say that having only sports examples is a bad idea? Explain briefly.”

Different people have different interests; sports is not liked by many, however, students tend to go toward the “easier” of the required courses. In this case, different themes of statistical studies may be more practical to peak interest for statistical studies for students (such as themes in biology, agriculture, business and many others).

Question 9

- (a) “Read the data into SAS and display the first 20 rows.”

The code ran to get the output:

```
filename myurl url "http://www.utsc.utoronto.ca/~butler/c32/footprint.txt";  
proc import
```

```

datafile=myurl
dbms=csv
out=footprint
replace;
getnames=yes;
delimiter=' ';

/* Viewing the data */
proc print data=footprint(obs=20);

```

The output is seen as follows

Obs	Country	Region	Eco_footprint
1	Argentina	A	2.5
2	Bolivia	A	2.1
3	Brazil	A	2.4
4	Chile	A	3
5	Colombia	A	1.8
6	CostaRica	A	2.3
7	Cuba	A	1.8
8	DominicanRep	A	1.5
9	Ecuador	A	2.2
10	ElSalvador	A	1.6
11	Guatemala	A	1.5
12	Haiti	A	0.5
13	Honduras	A	1.8
14	Jamaica	A	1.1
15	Mexico	A	3.4
16	Nicaragua	A	2
17	Panama	A	3.2
18	Paraguay	A	3.2
19	Peru	A	1.6
20	TrinidadToba	A	2.1

(b) “Make a suitable graph of the ecological footprint values for each Region”

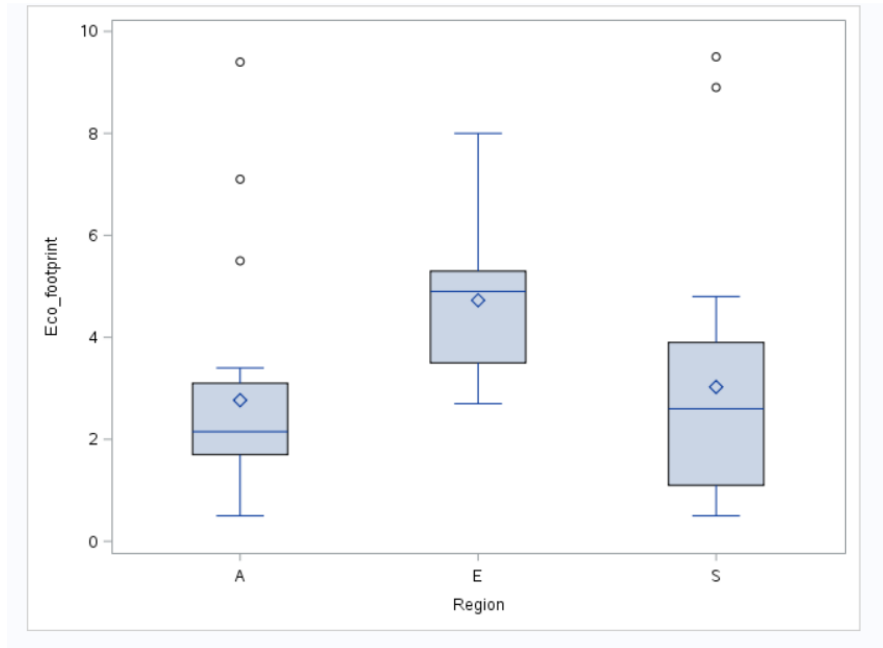
Since there is one categorical variable and one quantitative variable it is suitable to create a boxplot. The code is shown below:

```

proc sgplot;
  vbox eco_footprint / category=region;

```

The output of the code is:



(c) Find the mean and median ecological footprint by region.

The code to answer this question is:

```
proc means mean median;
  class Region;
```

The output is found as follows

The MEANS Procedure

Analysis Variable : Eco_footprint			
Region	N Obs	Mean	Median
A	24	2.7666667	2.1500000
E	23	4.7260870	4.9000000
S	19	3.0263158	2.6000000

By this table, we can see that region A has a mean of 2.67 and a median of 2.15. We see that region B has a mean of 4.726 and a median of 4.9. We see that the region S has a mean of 3.026 and a median of 2.6

(d) “Explain briefly how your graph and your calculations are consistent”

Though it would be nice to simply say “because SAS is programmed to do so”, there are other ways to say that the calculations are consistent.

Given the MEANS Procedure, we see the median and mean of each category. If we were to look at our boxplot, we see that the means for each group (the diamonds) are relatively around the same range as the MEANS procedure.

Likewise, the medians of each group in the boxplot (the middle line of each box), is relatively around the same range as the MEANS procedure.

Therefore, the calculations seem to be consistent.